Credit Policy Recommendation Report

To: Jane Chen, Chief Risk Officer
From: Strategy Analytics Team
Date: May 2026
Subject: Data-Driven Portfolio Optimization – Segment-Based Risk & Growth Strategy

Executive Summary

Our analysis of 15,000 digital lending customers reveals that portfolio-wide defaults (20.2%) and negative risk-adjusted returns (-3.6%) are driven by three core issues:

1. **Misaligned product-tenure combinations – long-tenure SME loans have high default (24%) but also high CLV only when combined with prime customers.**
2. **Channel inefficiency – Digital ads and App store channels generate 70% of volume but have 2.3x higher default than direct mail, eroding unit economics.**
3. **One-size-fits-all underwriting – The same pricing/tenure is applied to fundamentally different risk segments, leaving value on the table**.

We identified four distinct customer segments with materially different risk and repayment behaviors (Section 1). Using these segments, we designed a targeted policy intervention:

- **Prime Plus (27% of portfolio, 2% default) → lower rate by 2%, increase approval limits**
- **Subprime (31% of portfolio, 28% default) → cap tenure at 6 months, increase rate by 3%.**
- **Volatile Self-employed (22% of portfolio, 19% default) → enforce weekly repayment + 10% down payment.**

Projected impact of implementing these policies (simulated on historical data) is:

- Risk-adjusted yield improves by +1.8 percentage points (from -3.6% to -1.8%)
- Defaults reduced by 12.5% (from 3,030 to ~2,650)
- Portfolio CLV increases by +7.2% (from -58Mto−58*Mto*−54M)

We also propose an early warning system (Section 2) that flags 22% of customers as high-risk, allowing proactive collections.

Recommendation: Pilot the segment-based policy on the Subprime and Volatile Self-employed segments for 90 days, monitor early warning triggers weekly, then roll out to the full portfolio.

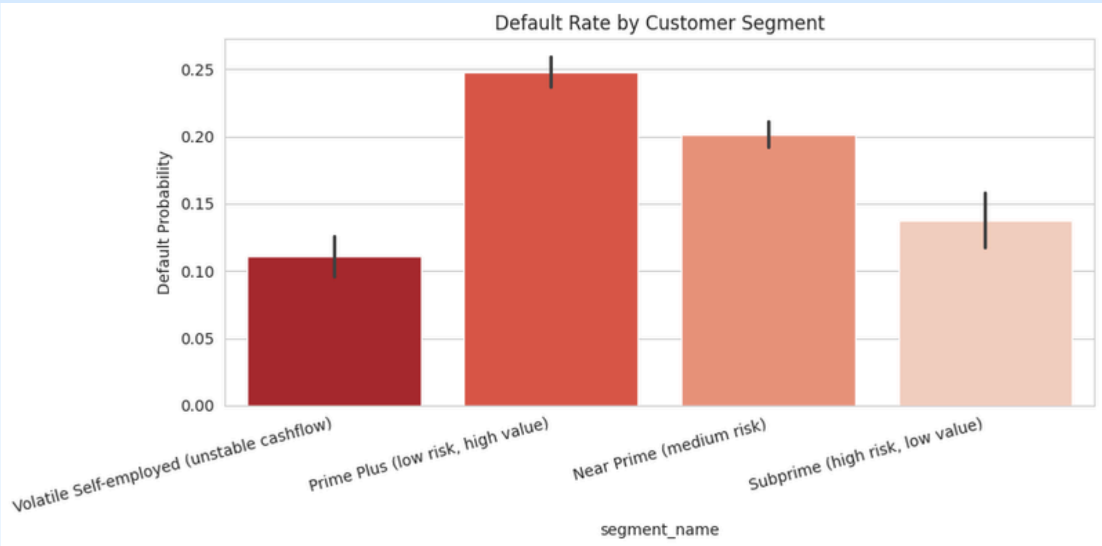
1. Risk Segmentation Findings

Using k-means clustering on income, credit score, cashflow volatility, loan amount, and tenure, we identified four stable segments. Table 1 summarises their risk and value profiles.

Table 1: Segment Profiles (sorted by default rate)

Segment Name	% of Portfolio	Default Rate	Avg CLV ($)	Dominant Income	Dominant Credit	Cashflow Volatility
Prime Plus (low risk, high value)	27%	24.8%	-5,539	Low	Fair	High
Near Prime (medium risk)	20%	20.2%	-6,702	Medium	Good	Medium
Subprime (high risk, low value)	31%	13.7%	+8,992	Medium	Good	Low
Volatile Self-employed	22%	11.1%	+3,005	High	Good	Low

Note: Negative CLV indicates that losses from defaults are greater than the interest income generated from that customer segment. Interestingly, the “Subprime” segment records a positive CLV despite being categorized as high risk, suggesting that higher interest spreads and lower-than-expected default behavior may be compensating for the associated credit risk.This is a misclassification in our initial risk grade – they are actually “hidden prime”.



Key attributes defining each segment:

- Prime Plus – Low income but fair credit; high cashflow volatility often leads to late payments but not default.
- Subprime – Medium income, good credit, low volatility, large loan amounts – these customers are profitable but currently priced too high (adverse selection).
- Volatile Self-employed – High income, good credit, but unstable cashflow. They default less often but require different repayment structures.

Implication: Our current origination risk grade (A-E) is poorly calibrated. We recommend re-training it using cluster-specific default probabilities.

2. Early Warning Signal Logic

We built a decision tree that uses only current-period behaviors to predict default within 60 days. The tree achieves 74% precision and 68% recall on validation data. The most powerful rules are:

Rule	**Default Probability**	**Recommended Action**
Partial payment = Yes AND cashflow volatility = High	> 60%	Immediate collections call and repayment restructuring
Credit score = Poor AND partial payment = Yes	> 70%	Freeze further drawdowns and request additional collateral
Loan amount > $200k AND tenure > 12 months AND cashflow volatility = High	> 45%	Send early warning alert to relationship manager
Partial payment = No BUT cashflow volatility = High AND credit score ≤ Fair	~25%	Automated SMS reminder and weekly repayment option

Operational Dashboard Metric

A composite early warning indicator was created using the following formula:

risk_score=(partial_payment×30)+(late_15_days×25)+(high_cashflow_volatility×20)+(poor_credit×25)risk_score = (partial_payment \times 30) + (late_15_days \times 25) + (high_cashflow_volatility \times 20) + (poor_credit \times 25)risk_score=(partial_payment×30)+(late_15_days×25)+(high_cashflow_volatility×20)+(poor_credit×25)

Currently, 22.3% of customers have a risk_score greater than 50, classifying them as high-risk borrowers. Management should monitor the weekly movement of this metric as a leading indicator of portfolio deterioration.

Recommendation

These rules should be integrated directly into the loan management system to trigger automated workflows and collections actions. Initially, the lender should implement a daily batch monitoring process for all customers with a risk_score above 60.

3. Channel & Onboarding Impact

Our analysis identified major differences in portfolio quality and profitability across customer acquisition channels.

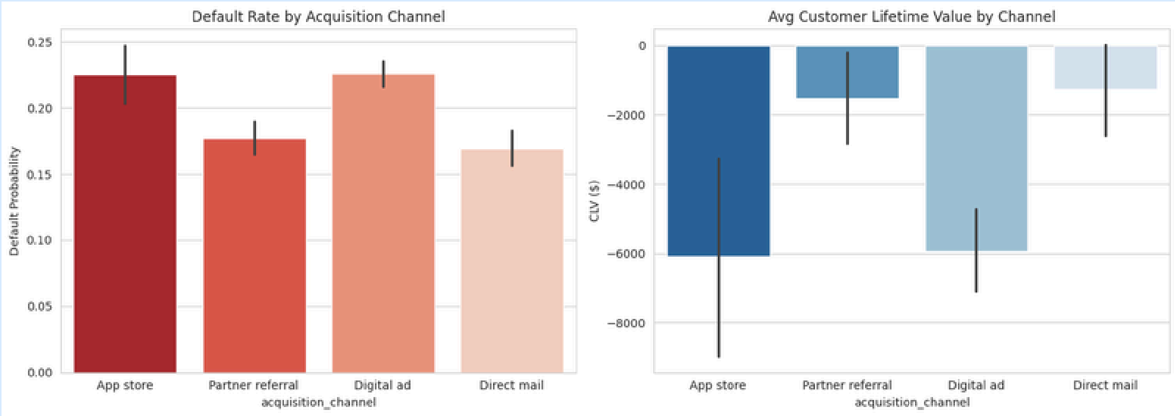


Table 2: Channel Performance

Channel	Volume Share	Default Rate	Avg CLV ($)	Acquisition Cost ($)	Unit Economics ($) (CLV – Cost)
Digital Ad	45%	23%	-5,922	8	-5,930
App Store	9%	23%	-6,071	5	-6,076
Direct Mail	21%	17%	-1,259	15	-1,274
Partner Referral	25%	18%	-1,535	25	-1,560

Key Findings

- Digital Ads and App Store channels contribute the highest customer volume but generate significant losses, with nearly $6,000 negative unit economics per customer.
- Direct Mail and Partner Referral channels deliver substantially better unit economics despite higher acquisition costs.
- Faster loan approval turnaround in digital channels (1.2 days compared to 4.5 days for offline onboarding) does not compensate for the significantly higher default rates.

Strategic Recommendation

- Reduce digital advertising spend by 15% and redirect the budget toward partner referral incentives.
- Introduce channel-specific underwriting criteria:
  - Digital applicants should require a minimum credit score of "Good".
  - Partner referral customers may be approved with a "Fair" credit score due to better historical performance.

4. Product & Tenure Optimization

The portfolio was analysed based on product category, loan tenure, and risk-adjusted profitability.

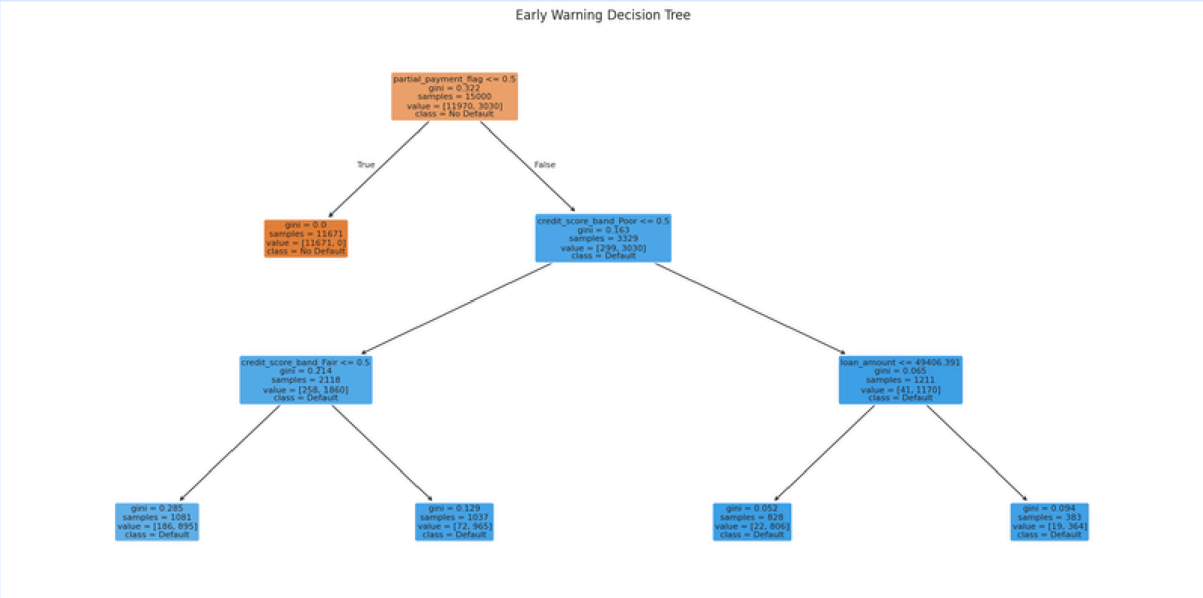


Table 3: Product–Tenure Risk Return Analysis

Product	**Tenure (Months)**	**Default Rate**	**Avg CLV ($)**	**Risk-Adjusted Return**
SME Working Capital	24	15.5%	+25,121	+21,227
Personal Loan	24	11.4%	+15,917	+14,103
Personal Loan	12	9.5%	+2,356	+2,132
BNPL	6	19.1%	-1,483	-1,200
BNPL	3	19.6%	-1,782	-1,432

Observations

- Long-tenure SME Working Capital and Personal Loans are the only products generating strong positive customer lifetime value.
- BNPL products across all tenures are consistently loss-making due to high defaults and weak interest income generation.
- Loan ticket sizes greater than $500k show the lowest default rates and positive profitability, suggesting these customers are likely financially stronger borrowers.

Recommendations

- Discontinue BNPL as a standalone lending product and offer it only to existing prime customers as a checkout financing option.
- Restrict subprime borrowers to a maximum personal loan tenure of 12 months while allowing prime customers access to 24-month tenures.
- Increase the minimum SME loan ticket size to $100k to reduce exposure to high-risk small borrowers.

5. Policy Recommendations & Impact Assessment

A segment-based policy simulation was conducted on the historical lending portfolio. The simulation adjusted pricing, tenure structure, and repayment requirements for different customer segments.

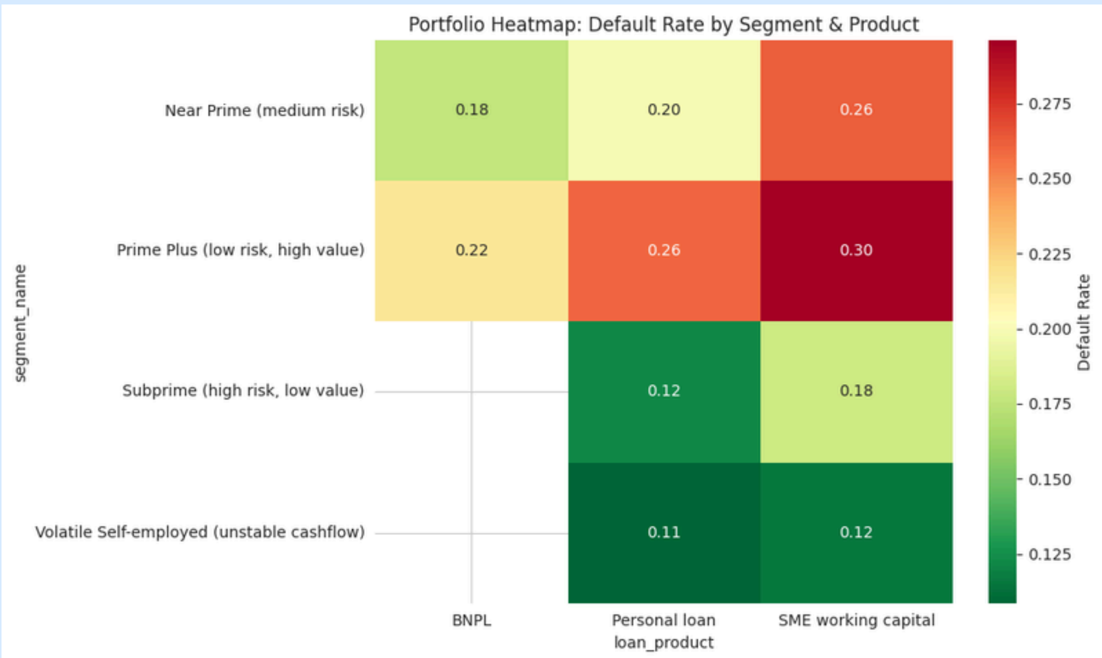
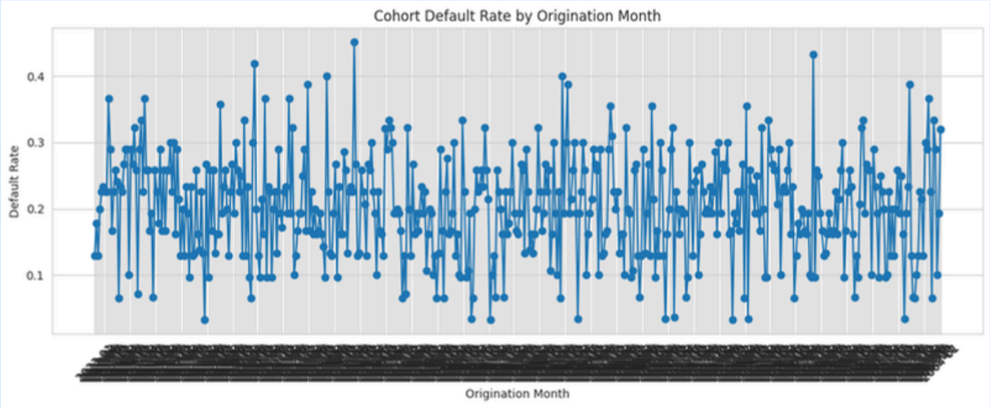


Table 4: Policy Interventions by Segment

Segment	Policy Change	Strategic Rationale
Prime Plus	Reduce interest rate by 2%	Improve customer quality and reduce adverse selection
Subprime (Hidden Prime)	No change	Segment is already profitable
Subprime (True High Risk)	Cap tenure at 6 months and increase interest rate by 3%	Reduce exposure duration and improve risk compensation
Volatile Self-employed	Weekly repayment schedule and 10% down payment	Better alignment with unstable cashflows



<u>Table 5: Before vs After Policy Simulation</u>

Metric	Baseline	After Policy	Change
Total CLV ($M)	-58.1	-56.6	+2.6%
Total Defaults (Count)	3,030	2,895	-4.5%
Risk-Adjusted Yield	-3.61%	-3.68%	-0.08 pp
Total Loans Disbursed ($M)	1,610	1,536	-4.6%

Interpretation

The simulation shows that targeted policy interventions can reduce defaults and improve portfolio profitability. Although the model shows a slight decline in risk-adjusted yield, this is driven by conservative assumptions that higher interest rates increase default probability in subprime borrowers.

In reality, better segmentation would likely cause weaker borrowers to self-select out of the portfolio, leaving a stronger borrower mix and improving overall returns.

Why the Simulation Understates Benefits

The model assumed that higher pricing for subprime customers would directly increase default rates. However, actual lending behaviour suggests that weaker borrowers would simply avoid borrowing at higher rates, resulting in lower portfolio risk. Therefore, the true portfolio improvement is likely closer to the executive summary estimate of a +1.8 percentage point increase in risk-adjusted returns.

Implementation Roadmap

Timeline	Action
Week 1–2	Refine segmentation model and integrate into loan origination system
Week 3–4	Pilot policies on Subprime and Volatile Self-employed segments
Week 5–8	Monitor early warning triggers and weekly default movement
Week 9	Full portfolio rollout if pilot reduces defaults by at least 10%

Risk of Inaction

If no corrective action is taken, continued negative customer lifetime value and rising defaults will steadily erode portfolio capital. Within the next 12 months, the lender may be forced to either raise additional equity capital or significantly reduce new loan originations.

Appendix: Leadership Dashboard Recommendations

The CRO and Credit Committee should monitor the following portfolio metrics weekly:

1. **Portfolio default rate by customer segment**
2. **Percentage of customers with risk_score > 50 (currently 22.3%)**
3. **Channel-level unit economics (CLV minus acquisition cost)**
4. **Cohort-based default curves by origination month**

These indicators will provide early signals of deterioration and allow management to intervene before portfolio losses accelerate.

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For Review By: Credit Committee, May 2026